Debjani Nath Sarma

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PROFESSIONAL SUMMARY

Data Analyst with 8+ years across retail, banking, and pharmaceutical/healthcare. Strengths in SQL, Python, Snowflake and Power BI. Built dashboards and ML models to translate complex requirements into data-driven solutions and actionable insights.

SKILLS

Data Analytics and BI: SQL(SSMS, MySQL, Oracle DB), Python(Pandas, NumPy, Matplotlib), Power BI(DAX, Power Query), Tableau, Excel(Pivot Table, XLOOKUP, VLOOKUP), R(tidyverse)

Data Engineering: Data Integration(dbt, ETL/ELT), Data Modeling, Data Warehousing(Snowflake, Redshift), Data Wrangling

Statistics and Modeling: A/B Testing, Hypothesis Testing, Regression Analysis, Time Series Analysis, Correlation Analysis, Predictive Modeling, Classification and Clustering, Explainable AI(SHAP)

Project Management: JIRA, Agile, Scrum, Stakeholder Collaboration, UAT Coordination, Requirements Gathering

PROFESSIONAL EXPERIENCE

Research Data Analyst, University of South Florida - Tampa, FL | Jan 2026 - Present

- Analyzed 150K+ CHF inpatient admissions across 4K+ U.S. hospitals using Python/pandas and developed a multidimensional hospital financial risk index (COFRI), enabling data-driven evaluation of clinical, operational, and financial variables.
- Developed machine learning classification models on HCUP NIS datasets to execute Predictive Modeling and Advanced Analytics, achieving a 0.769 AUC to flag high-risk facilities and enable proactive intervention targeting.
- Identified emergency admission rate, patient severity, and diagnosis count as top financial-risk predictors using Explainable AI (SHAP), reframing financial risk as a downstream signal of upstream clinical and operational patterns.

Graduate Teaching Assistant, University of South Florida - Tampa, FL | Jun 2025 - Dec 2025

- Mentored 130+ graduate students in SQL, Python, Tableau, and Power BI as a teaching assistant across multiple business analytics courses.
- Built hands-on practice exercises and delivered 1-on-1 debugging support to bridge the gap between lecture concepts and applied problem-solving.

Senior Data Analyst, Accenture - Bengaluru, India | Jan 2022 - May 2023

- Collaborated with cross-functional stakeholders for a global fashion retailer to architect a unified data flow, driving consistency across 3+ downstream reports.
- Analyzed historical sales and product performance to identify top-selling categories and seasonal trends, informing inventory planning decisions reducing stockouts by 15%.
- Led root cause analysis of data discrepancies between POS and inventory systems, improving data integrity and reporting reliability among 20+ users.
- Conducted UAT and reporting validation, ensuring analytics solutions are aligned with business requirements and reducing reporting defects by 30%.

Associate Analyst, Software AG - Bengaluru, India | Mar 2020 - Dec 2021

- Developed RESTful API pipelines to streamline financial transaction data for real-time compliance reporting and auditing at a Tier-1 global bank.
- Led a team of 4 engineers to deliver robust integration modules on schedule, coordinating sprint planning, task allocation and progress tracking using JIRA to achieve key project milestones.
- Drove cross-functional UAT and go-live coordination with compliance stakeholders, ensuring full test coverage and defect resolution, resulting in real-time transaction visibility and a centralized audit trail.

Analyst - Data and Integration, Accenture - Bengaluru, India | Mar 2016 - Feb 2020

- Analyzed enterprise data flows across SAP HANA, IBM Maximo, Ariba, and CRM platforms across clients in entertainment, energy, pharmaceutical, and utilities sectors; partnered with product owners, business analysts and leads to translating business requirements into technical data specifications powering integration workflows.
- Wrote SQL queries on business data to validate data integrity, generate recurring operational reports, and investigate data discrepancies; surfaced error patterns that prevented 15+ defects across multiple release cycles.

- Designed automated error-alert services that proactively notified support teams of data and integration failures, reducing daily manual log monitoring by an estimated 30+ minutes per team member and improving incident response times.
- Coordinated UAT and defect tracking across projects, mentoring 3+ newer analysts on data validation, SQL querying, and testing practices, while building reusable Excel-based validation templates adopted by the team.

EDUCATION

Master of Science, Business Analytics | Aug 2024 - Dec 2025
University of South Florida - Florida, USA

Bachelor of Technology, Computer Science and Engineering | Jul 2015
Sikkim Manipal University - Sikkim, India

PROJECT EXPERIENCE

U.S. Crime & Socioeconomic Trends Analysis (Python, Tableau) | Dec 2025

- Analyzed FBI UCR and U.S. Census data (2010-2019) across 50 states using Python for data cleaning, normalization, and feature engineering to enable consistent cross-state and temporal analysis.
- Built interactive Tableau dashboards visualizing socioeconomic factors against crime trends; identified that property crime correlates strongly with low-income states while unemployment shows weak correlation with violent crime.

Movie Investment Analytics & ROI Prediction (Oracle SQL, Tableau, Python) | Jun 2025

- Designed a multi-dimensional data warehouse and built OLAP schemas for 45,000+ movies and 20M+ user ratings to perform SQL analysis based on users, time-period, region, genre ROI, audience sentiment, and release trends.
- Developed interactive Tableau dashboard to visualize genre performance, ROI, and budget-to-revenue relationships while integrating Random Forest model insights to identify high-ROI genres such as Crime and Comedy, enabling data-driven film investment and content strategy decisions.

Airline Customer Retention (Python, Predictive Modeling) | Jan 2025

- Performed feature engineering, data preprocessing and exploratory data analysis on an airline customer survey dataset with 100,000+ records using Pandas, NumPy and Matplotlib to detect patterns and trends.
- Designed Random Forest model with 96% accuracy and an XG Boost model with 95% accuracy using Python to identify key factors influencing customer retention in the airline industry.